Display tech

Display panels, big and small are popping up wherever they can be placed. Here is a little know-how about all the technologies behind these panels

CRT

Cathode Ray Tube is the oldest of all display technologies that relies on a vacuum tube in which an electron gun fires electrons onto a fluorescent screen to project the image. They are bulky and expensive to produce compared to more popular technologies used today. CRTs were produced with the use of multi-coloured phosphorous colour.

LCD/TFT

They are one and the same and utilize thin transistors which align whenever electricity is passed across them. This lets light pass through the screen. The backlight for these panels may be provided by CFL tubes or LEDs. LEDs have better black levels and low energy consumption. There are multiple types of LCD/TFT panels:

- **TN**

Twisted Nematic Film – one of the earliest and cheapest type of LCD panels out there. Each pixel has a helical arrangement of electrodes between two panels which passes light when switched off. When on, it blocks all light. They are characterised by restricted viewing angles, reduced colour gamut, 6-bit colour depth and fast response times.

- **IPS**

In Plane Switching – developed to overcome the drawbacks of TN panels, IPS panels had all electrodes parallel to the plane rather than in a helical format. They are characterized by wider viewing angles, increased colour gamut, 8-bit colour depth and reduced response time.

- **VA**

Vertical Alignment – It was brought out as a compromise between IPS and TN. Thus taking a hit in response time and colour-depth while having a wider viewing angle.

- **PLS**

Plane Line Switching – a recent development which would fit in between VA and e-IPS. It boasts of better viewing angles, low response time and comes with low contrast ratio.

- **Blue Phase**

Blue Phase Mode LCDs have very high refresh rates and subsequently low response time (1 microsecond is the least so far). Panels with refresh rates of 240 Hz have been manufactured.

Plasma

Plasma displays consist of miniature cubes filled with inert gases and mercury that when struck by electricity light up the respective phosphorous coated cube to light up a pixel. Since there is no light when the cells are off, the blacks produced by plasmas were considered to be the best. A major disadvantage to plasma displays are that if an image is displayed for long durations on the screen then it'll create a "burn", i.e. a watermark like appearance will permanently be present on the screen.

OLED

Organic Light Emitting Diode – these are organic semiconductors which give off light when a charge is passed across them. They are less power consuming, can be printed onto substrates. Subsequently they are used to create flexible displays with a response time as low as 0.1 microsecond. Also, they have wider viewing angles.

- **AMOLED**

Active Matrix OLED – are used more commonly to create panels for mobile displays but can be scaled up easily to produce large format displays. Active matrix displays have the ability to switch on individual elements for each pixel.

- **Super AMOLED**

This term is more of a marketing jargon for AMOLED panels which have a digitiser built into them rather than being an overlay.

e-paper

- **Gyricon**

These displays consist of individual charged particles which are dark on one side and light on the other, suspended in oil. They are embedded into a thin plastic sheet sandwiched between a pair of electrodes. When an electric charge is applied to electrodes on either side of the particle, it aligns itself. The charge need only be applied once and the particles will stay that way till realigned.

- **Electrophorectic**

Similar to gyricon, electrophoretic displays have small capsules filled with ink and titanium dioxide particles, which are similarly embedded in a plastic substrate. The smaller the capsule, the better the resolution. Most e-book readers in the market today use this technology.

Foldable displays

When implemented as a screen, the display surface is tracked by an infrared camera to map out the surface. Once done, visible light is projected onto the surface to create the image.

Then there are panels which are made out of organic TFT which allow for thin displays that can be rolled up or laid out on any surface to make it a display.

DLP

Digital Light Processing – uses a matrix of microscopic mirrors laid across a semiconductor chip (Digital Micromirror Device). Each individual mirror is triggered to reflect light through the projector lens or onto a light dump. Each mirror or each pixel can produce 1024 shades of grey. This DMD is then synchronized with a colour wheel which rotates really fast in front of the DMD. Hence, each shade of grey is converted into a shade of colour.

Parameters

- **Response time**

The time taken for a pixel to change from one state to another. Measured commonly as (Tr+Tf) where it's the total time taken for a pixel to change from black->white->black. The lesser the better.

- **Pixel pitch**

Refers to the distance between pixels. The lesser the PP, the smaller and sharper the image.

- **Colour Gamut**

Refers to the range of colours a display can produce, this has more to do with the backlight of the panel.

- **Refresh rate**

Tells you how fast the screen can draw an image. On a CRT monitor, it says how fast the cathode ray gun can refresh the entire image.